Exp. No: 2

Electric Field and Electric Flux Density in a Sphere Full of Charges with Radius Varying

Aim:

To write and validate MATLAB programs for electric field intensity (E) and electric flux density (D) for specified spherical charge distributions.

Objectives:

1. To derive expressions for electric field intensity (E) and electric flux density (D) inside and outside a uniformly charged sphere using Gauss's law.

2. To plot the variation of E and D with radial distance for varying sphere radius and charge density.

3. To validate the simulated results against theoretical calculations for different radii.

Test case (i) Determination and plotting of electric field and electric flux density in sphere full of charges with varying radius

```
clc;

clear;

close all;


% Input values

q = input('Enter the value of q = ');

r = input('Enter the value of r = ');


% Constants

epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);


% Calculate electric field and flux density
```

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```
E = k*q/(r^2);
D = epsilon*E;

% Display calculated values
disp('The electric field is:');
disp(E);

disp('The electric flux density is:');
disp(D);

% Plot graphs for varying radius
x = linspace(0.1, 10, 500);

E_plot = k*q./(x.^2);
D_plot = epsilon.*E_plot;

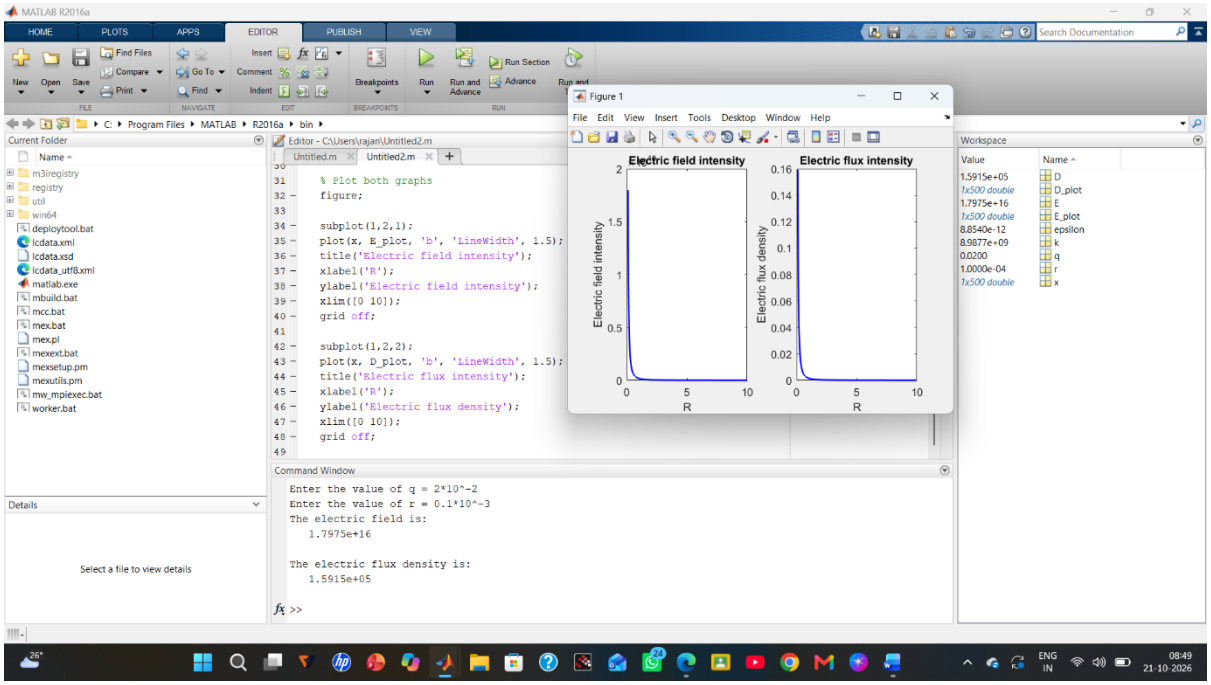
% Plot both graphs
figure;

subplot(1,2,1);
plot(x, E_plot, 'b', 'LineWidth', 1.5);
title('Electric field intensity');
xlabel('R');
ylabel('Electric field intensity');
xlim([0 10]);
```

```
grid off;


subplot(1,2,2);
plot(x, D_plot, 'b', 'LineWidth', 1.5);
title('Electric flux intensity');
xlabel('R');
ylabel('Electric flux density');
xlim([0 10]);
grid off;
```

Output:- Enter the value of q=2*10^-2 Enter the value of r=0.1*10^-3



Test case (ii) Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0

```
clc;
clear;
```

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```
close all;


rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;


r = linspace(0, R, 100);

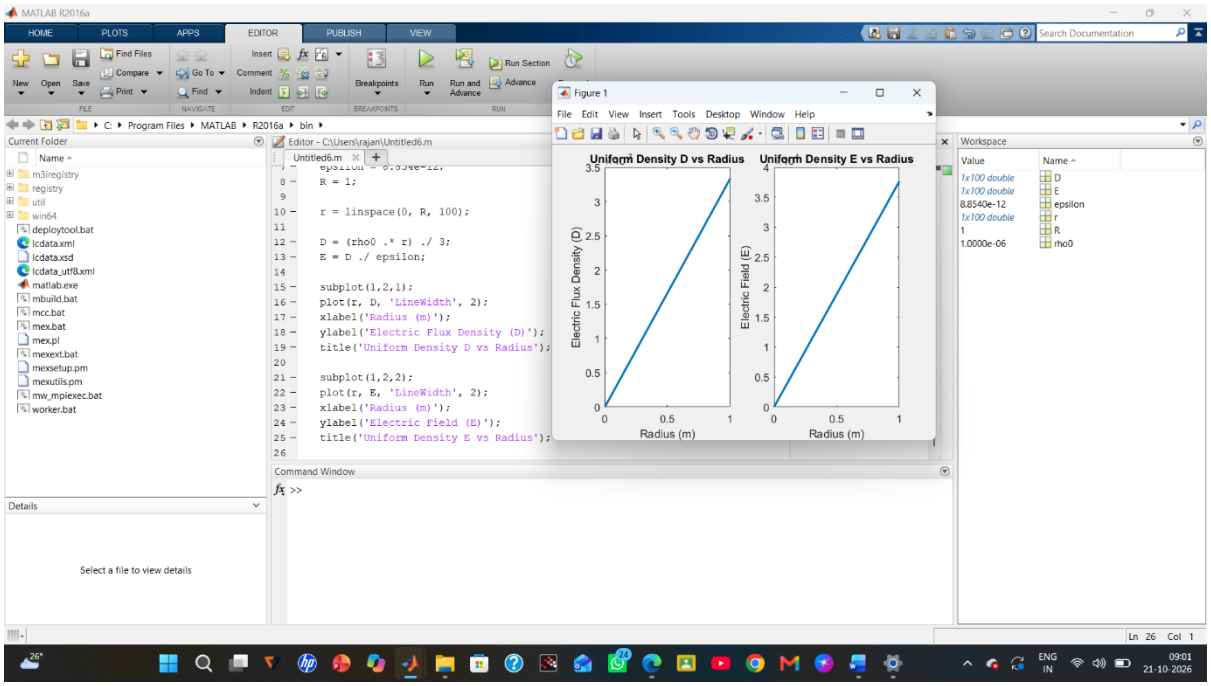

D = (rho0 .* r) ./ 3;
E = D ./ epsilon;


subplot(1,2,1);
plot(r, D, 'LineWidth', 2);
xlabel('Radius (m)');
ylabel('Electric Flux Density (D)');
title('Uniform Density D vs Radius');


subplot(1,2,2);
plot(r, E, 'LineWidth', 2);
xlabel('Radius (m)');
ylabel('Electric Field (E)');
title('Uniform Density E vs Radius');
```

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Test case (iii) For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive and plot the electric field (E) and electric flux density (D)

```
clc;
clear;
close all;

rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;

r = linspace(0, R, 100);

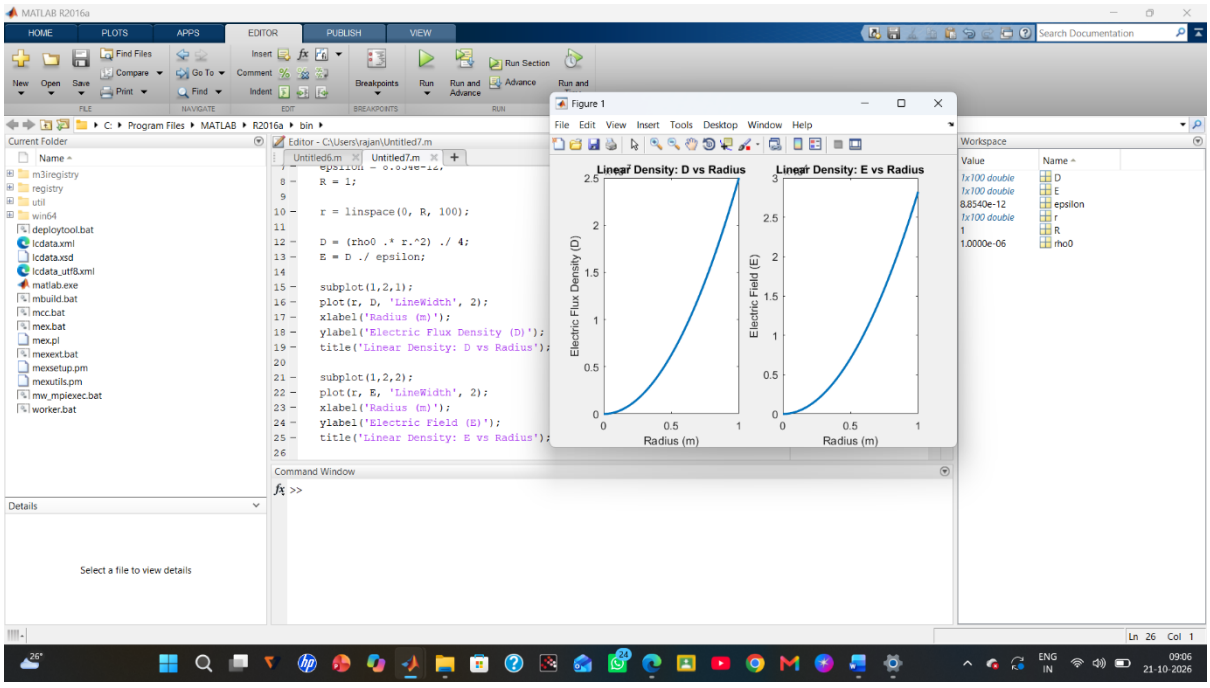
D = (rho0 .* r.^2) ./ 4;
E = D ./ epsilon;
```

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```
subplot(1,2,1);
plot(r, D, 'LineWidth', 2);
xlabel('Radius (m)');
ylabel('Electric Flux Density (D)');
title('Linear Density: D vs Radius');

subplot(1,2,2);
plot(r, E, 'LineWidth', 2);
xlabel('Radius (m)');
ylabel('Electric Field (E)');
title('Linear Density: E vs Radius');
```



Test case (iv): For a uniformly charged solid sphere of radius R = 2 m and volume charge density $\rho_v = 3\times10^{-6}$ C/m³, calculate and plot E and D inside and outside the sphere.

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```
clc;
clear;
close all;

rho0 = 3e-6;
epsilon0 = 8.854187817e-12;
R = 2;

r = linspace(0.4, 10, 401);
E = zeros(size(r));

inside = (r <= R);
outside = (r > R);

% Electric field inside the sphere
E(inside) = rho0 .* r(inside) ./ (3 * epsilon0);

% Electric field outside the sphere
E(outside) = rho0 * R^3 ./ ...
    (3 * epsilon0 .* r(outside).^2);

% Plot a single graph
figure;
plot(r, E, 'LineWidth', 1.5);
grid on;
```

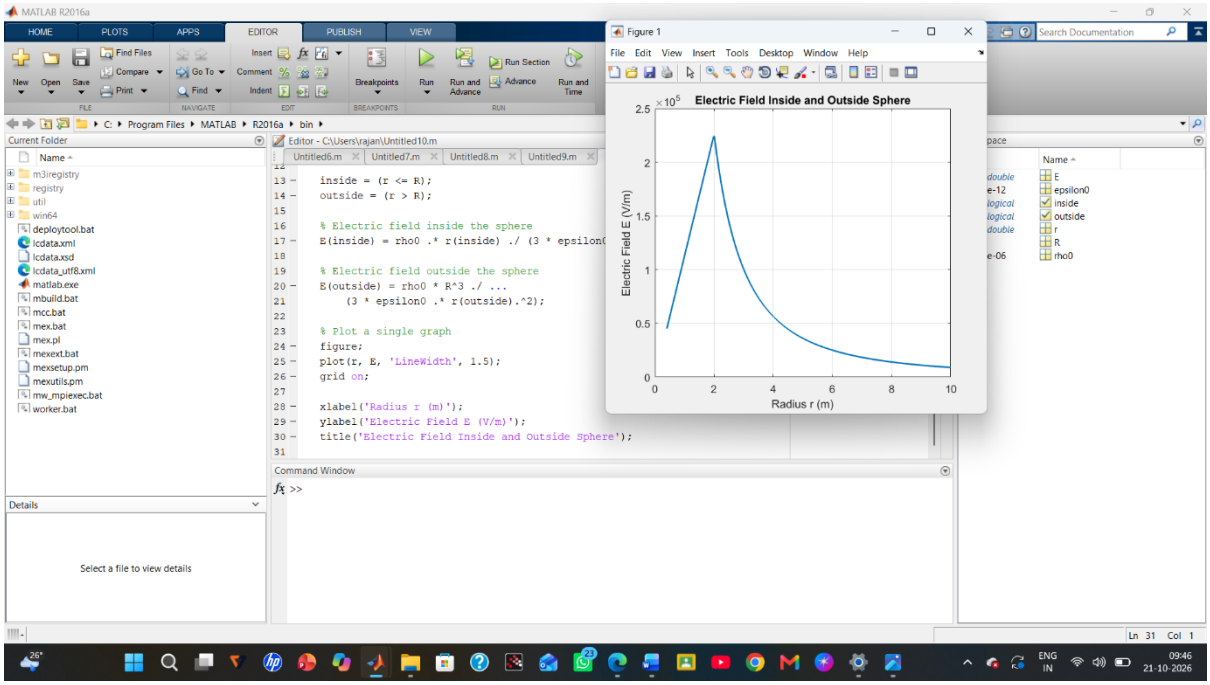
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xlabel('Radius r (m)');

ylabel('Electric Field E (V/m)');

title('Electric Field Inside and Outside Sphere');



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)
1	Select the appropriat e charge distributio n model	Select the appropriate charge distribution model	. Correctly distinguishes point, uniform volume and specified non uniform charge models and their assumptions.	Selects the correct model with minor clarifica tion needed.	Shows partial understandin g of the charge model..	Uses an unsuitable model or assumptions.	25
2	Implement E and D calculation s in MATLAB	Implement E and D calculation s in MATLAB	Correctly calculates electric field E and flux density D with	Calculat ions are mostly correct with minor coding or unit errors.	Some calculations work but require assistance.	Unable to obtain valid E and D values.	25

			consistent SI units.				
3	Analyze radial variation and validate results	Analyze radial variation and validate results	Plots the correct inside/outside or stated distribution behaviour and verifies against theory.	Plots and validation are mostly correct with minor discrepancies.	Plots or theoretical validation are incomplete.	Plots are incorrect or results are not validated.	30
4	Interpret and document findings	Interpret and document findings	Explains the effect of radius and charge distribution and presents labelled plots/tables.	Interpretation is generally correct with minor omissions.	Interpretation is limited or documentation incomplete.	Cannot interpret or document results.	20
Total							100